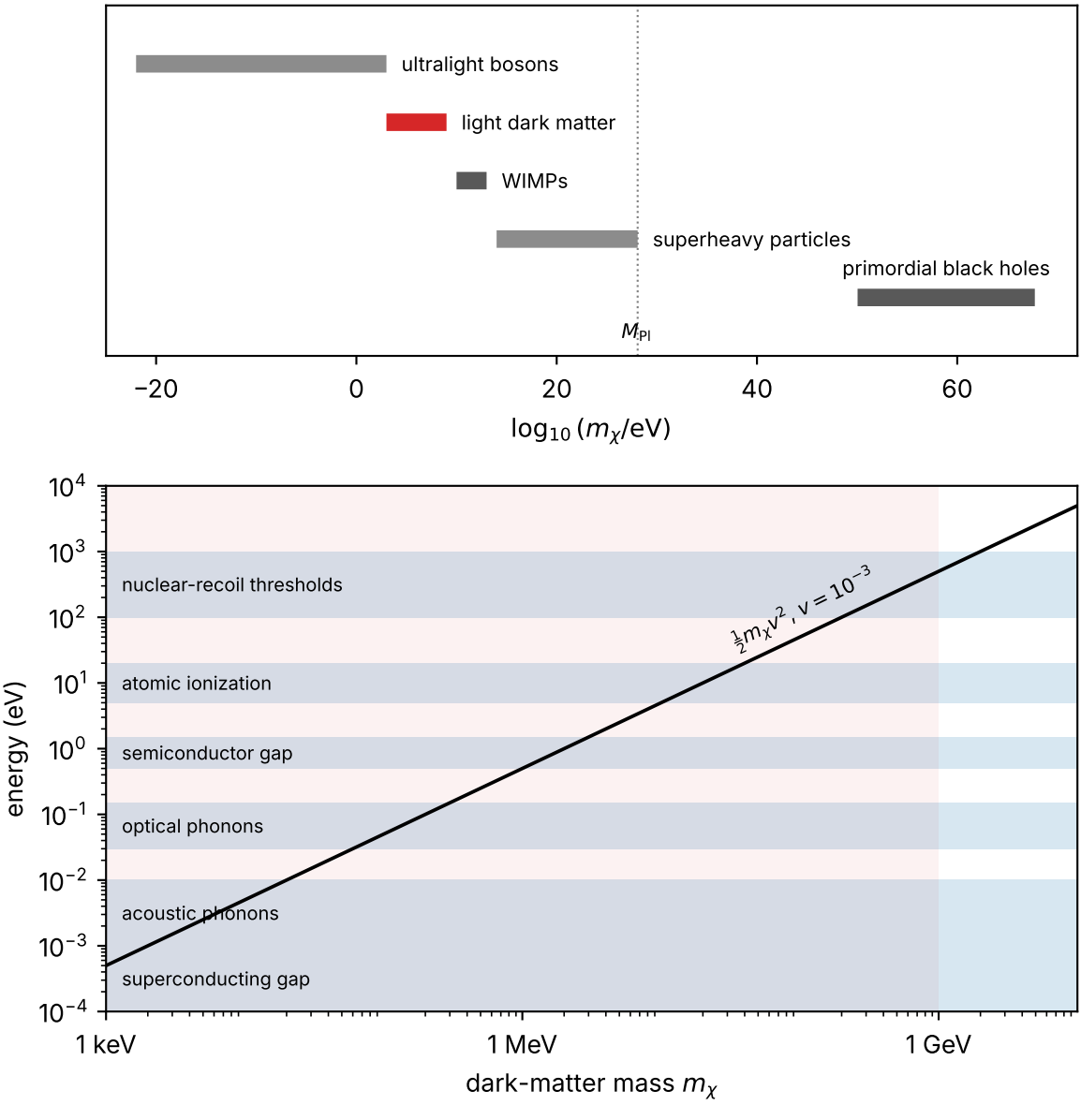
ultralight bosons
light dark matter
WIMPs
superheavy particles
primordial black holes
$M_{\rm Pl}$
−20
0
20
40
60
$\log_{10}(m_\chi/\mathrm{eV})$
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
energy (eV)
nuclear-recoil thresholds
atomic ionization
semiconductor gap
optical phonons
acoustic phonons
superconducting gap
$\frac{1}{2}m_\chi v^2$, $v = 10^{-3}$
1 keV
1 MeV
1 GeV
dark-matter mass m_χ